

INCIDENT REPORT — RIVAGE PM5 DSP ENGINE FAILURES

STEPHENSON THEATRE | 2026-07-09 (x2) | DSP-RX-EX | ERR X16 | rev. 1.1 — 2026-07-10 (supersedes 2026-07-09)

Summary

During tech week for *Les Misérables*, the Yamaha RIVAGE PM5 system at the Stephenson Theatre suffered **two spontaneous DSP engine failures on 2026-07-09** — at 13:29:54 (mid 10-out-of-12) and again at 22:31:07 — with identical signatures: the control surface logged "DSP disconnected!" and lost all 21 networked I/O devices simultaneously; the DSP-RX-EX engine froze hard, logging no fault, shutdown, or restart of its own. After each reset the engine's front panel displays **[ERR X16] MEMORY ERROR** (red, flashing).

The root cause is under active joint investigation with Yamaha Technical Support (ticket opened 2026-07-10; Clearwing copied). The on-panel error code is classified by the RIVAGE PM Operation Manual as an internal-component fault ("One of the internal parts is faulty... Repair might be required. Contact your Yamaha dealer."). Yamaha, however, has not confirmed a hardware fault and is also examining the system's PC-network configuration as a possible factor. **Neither explanation has been confirmed or excluded at this time.** The failure has occurred on two firmware versions (V6.71 previously; V7.00 both times on 07-09), ruling out the 2026-07-02 firmware update as the cause.

The show opens **July 17, 2026**, and the system has no mirrored/redundant DSP engine. An expedited resolution is requested before opening — whether that is repair/loaner (if hardware is confirmed) or a validated configuration remedy (if not).

System

Console	Yamaha RIVAGE PM5 (CS-R5) + DSP-RX-EX, Theatre Mode, 96 kHz
Firmware	DSP-RX-EX V7.00 (updated 2026-07-02 from V6.71; both versions have exhibited the failure)
DSP	Unit ID 1 · engine IP 10.18.11.40 /24 · surface IP 10.18.11.49 /24
PC Network	Both CONSOLE and DSP ENGINE IPs enabled/static on the control VLAN. Physically, only the CS-R5 PC port is connected; the engine PC port is unconnected (verified 2026-07-10). See Observation 7.
HY cards	HY1 96k I/O (10.18.10.28), HY3 SRC (.27), HY4 records (.33), Dante primary/secondary networks
I/O at failure	21 devices incl. Rio3224-D (Y005), Rio3224-D2 x2 (Y007/Y008), Rio1608-D2 (Y009)

Timeline (from system log exports of 2026-07-09 13:32 and 2026-07-10 09:25)

2026-07-02 00:59	DSP-RX-EX firmware updated V6.71 → V7.00 ("V7.00 Program Start")
2026-07-08 19:25	Single I/O dropout (HY SLOT1 device #20), self-recovered in 26 s
2026-07-09 13:29:54	FAILURE #1 — CS: WARNING "DSP disconnected!"; all 21 I/O devices lost simultaneously (HY slots 1/3/4)
2026-07-09 13:30:55	Show file saved [Les Mis 7.09.RIVAGEPM] (surface still up); logs exported 13:32
2026-07-09 13:38:50	DSP restarted — came up in Unit Mode: Default (not Theatre); "Cannot Sync: UNIT MODEs are different" until corrected 13:39:55
2026-07-09 13:39:57	DSP logged Cooling Fan State: [Error] — cleared on the subsequent 13:40:53 restart (Theatre mode)
2026-07-09 22:31:07	FAILURE #2 — identical signature: "DSP disconnected!", all I/O no-response (Y005/Y007/Y008/Y009)
2026-07-09 22:42:06	DSP restarted (Theatre mode); post-reset front panel again [ERR X16] MEMORY ERROR
2026-07-10 09:25	Second log set exported; logs from both failures delivered to Yamaha

Observations (facts on record; interpretation pending the joint investigation)

- 1. The engine logs nothing at failure time.** Both failures: no error, shutdown, or restart entry from the DSP itself; the surface loses the engine and every HY-slot I/O device in the same second.
- 2. ERR X16 after both failures.** The manual classifies this code as an internal-component fault ("Repair might be required"). Yamaha has not yet confirmed that reading against the logs.
- 3. Not the firmware.** Same failure previously under V6.71 (department records; predates the exported log history) and twice under V7.00.

4. Power ruled out. All logged power-state events (2026-01-06, 02-18, 07-02) were confirmed manual shutdowns by staff; both supplies feed from a power conditioner. No unexplained power events precede either failure.

5. The Dante layer stayed up through failure #2. Dante Controller remained green — the HY cards' network interfaces kept running while the surface showed the engine as disconnected, so the chassis had power at the moment of failure.

6. Post-failure-#1 anomalies. The engine rebooted into Unit Mode Default rather than Theatre, and logged a momentary cooling-fan error minutes after restart. Significance not yet established.

7. PC-network observation (under review with Yamaha). The DSP ENGINE IP is enabled and statically addressed although the engine's PC port is not physically connected. A VLAN 11 packet capture (2026-07-07, two days before the failures) shows the engine's MAC active on the control network regardless — as both its static address and a second, factory-default subnet (192.168.53.x) — reaching the wire via the console's single PC-port connection. Yamaha is evaluating whether this configuration is a factor. The capture file has been offered to Yamaha.

Status and pending actions

Yamaha ticket active (2026-07-10); both log sets delivered; Clearwing copied at our request. In progress on the department side, per Yamaha's direction: uploading a fresh developer log (*dvlp.log); disabling the unused DSP ENGINE IP; verifying nothing is connected to the network port above the HY1 slot; and correcting a word-clock configuration item Yamaha flagged as improper though likely unrelated to these failures (internal clock selected while no Dante card is set to external sync and all three cards are set Preferred Leader).

Request

Expedited resolution before opening night **2026-07-17** (performance schedule Wed–Sun through 2026-10-18): repair or loaner DSP-RX-EX if the hardware fault is confirmed, or a validated configuration remedy if the investigation lands there. Full system logs, the post-crash show file, and the VLAN 11 packet capture are available on request. Department contact: Matthew Gibney, Audio Supervisor.